2.3 Group 17

Question Paper

Course	CIE A Level Chemistry
Section	2. Inorganic Chemistry
Topic	2.3 Group 17
Difficulty	Medium

Time allowed: 20

Score: /10

Percentage: /100

Question 1

On being heated, hydrogen iodide breaks down more quickly than hydrogen chloride.

Which statements explain this faster rate?

1. The breakdown of HCl is more exothermic than that of HI.
2. The reaction of the breakdown of HI has smaller activation energy than that of HCl.
3. The HCl bond is stronger than the HI bond.

A. 1 only

B. 1 and 2

C. 2 and 3

D. 1, 2 and 3

[1 mark]

Question 2

The following report appeared in a newspaper:

> Barrels of bromine broke open after a vehicle collision on the motorway. Traffic was diverted as purple gaseous bromine drifted over the surface of the road (as bromine is more dense than air), causing irritation to drivers' eyes. Firemen sprayed water over the scene of the accident, dissolving the bromine and washing it away.

Which of the following observations is **incorrect** in the report?

A. Bromine is not purple

B. Bromine does not vaporise readily

C. Bromine is not denser than air

D. Bromine does not dissolve in water

[1 mark]

Question 3

What is the complete list of all the products from the reaction of potassium bromide with concentrated sulfuric acid?

A. Potassium hydrogen sulfate, hydrogen bromide, bromine, water and sulfur dioxide

B. Potassium hydrogen sulfate, hydrogen bromide, bromine and water

C. Potassium hydrogen sulfate, hydrogen bromide and bromine

D. Potassium hydrogen sulfate and hydrogen bromide

[1 mark]

Question 4

On contact with a hot glass rod, which gaseous hydride most readily decomposes into its elements?

A. Ammonia

B. Steam

C. Hydrogen iodide

D. Hydrogen chloride

[1 mark]

Question 5

What happens when chlorine is bubbled through an aqueous sodium hydroxide solution?

1 In hot NaOH (aq), ClO_3^- (aq) ions are formed.

2 In cold NaOH (aq), ClO^- (aq) ions are formed.

3 Disproportionation of chlorine occurs in both cold and hot aqueous solutions.

A. 1 only

B. 1 and 2

C. 2 and 3

D. 1, 2 and 3

[1 mark]

Question 6

A molecule of chlorine has a relative molecular mass of 72.

Which properties of the atoms in this molecule are the same?

1 Radius

2 Relative isotopic mass

3 Nucleon number

A. 1 only

B. 1 and 2

C. 2 and 3

D. 1, 2 and 3

[1 mark]

Question 7

Astatine is the element below iodine in Group 17 of the Periodic Table.

Which statement is most likely to be true for astatine?

A. Silver astatide reacts with dilute aqueous ammonia in excess to form a solution of a soluble complex.

B. Sodium astatide and hot concentrated sulfuric acid react to form astatine.

C. Astatine and aqueous potassium chloride react to form aqueous potassium astatide and chlorine.

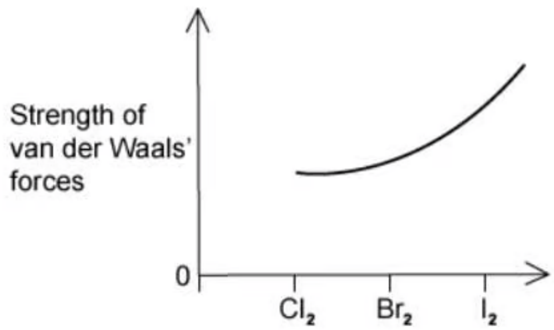
D. Potassium astatide and hot dilute sulfuric acid react to form white fumes of only hydrogen astatide.

[1 mark]

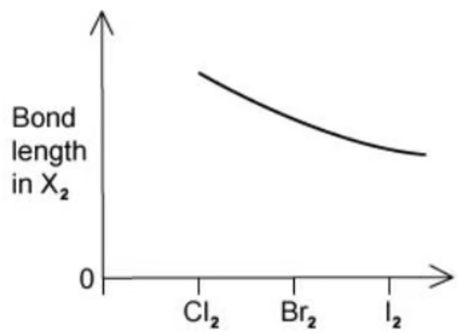
Question 8

Which graph correctly illustrates a trend found in the halogen group?

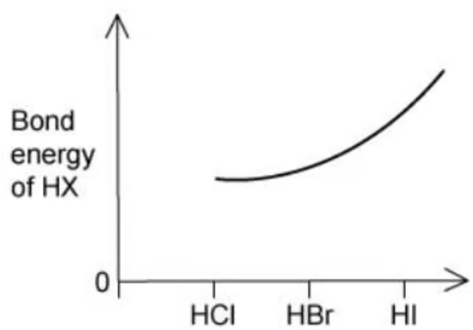
A.



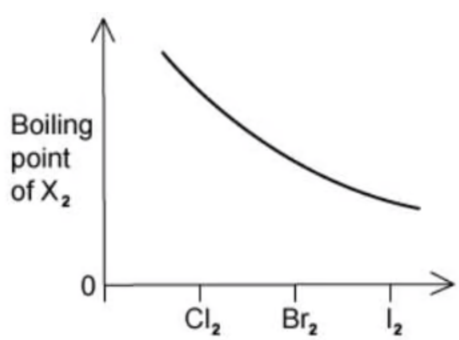
B.



C.



D.



[1 mark]

Question 9

When the yellow liquid NCl_3 is stirred into aqueous sodium hydroxide, the reaction that occurs can be represented by the following equation.

$$2NCl_3\,(l) + 6NaOH\,(aq) \rightarrow N_2\,(g) + 3NaCl\,(aq) + 3NaOCl\,(aq) + 3H_2O\,(l)$$

What will be the result of this reaction?

1 A bleaching solution remains after the reaction.

2 The nitrogen is oxidised.

3 The final solution gives a precipitate with acidified silver nitrate.

A. 1, 2 and 3

B. 1 and 3

C. 2 and 3

D. 1 only

[1 mark]

Question 10

Compound **X** readily conducts electricity when added to water to produce a solution but does not conduct electricity in its liquid state.

Which of the following could **X** be?

1 $MgCl_2$

2 PCl_3

3 $SiCl_4$

A. 1, 2 and 3

B. 1 and 2

C. 2 and 3

D. 1 only

[1 mark]

Model Answers: Medium

1

The correct answer is **C** because:

- The thermal stability of the hydrogen halides decreases as you go down the group.
- This is because the hydrogen halide bond gets weaker, so the H*Cl* bond is stronger than the HI bond.
- The **activation energy** of HI is less than the activation energy of H*Cl* so; therefore, more partic*l*es will have at least the activation energy when heated.

Statement 1 is incorrect as thermal decomposition is an endothermic reaction and requires the input of energy.

2

The correct option is **A** because:

- The question is asking you to read the report and pick which fact stated in the report is incorrect.
- The report states the bromine gas was purple, this is wrong because **bromine gas is not purple.**
- Bromine gas is **orange/brown.**
- The colour of the gases of Group 17 elements:
    - Fluorine – pale yellow
    - Chlorine – green/yellow
    - Bromine – orange/brown
    - Iodine – black solid, purple vapour

B is incorrect as the report suggests that bromine vaporises easily (as the bromine was gaseous), this is true – **bromine vaporises readily.**

C is incorrect as the report states that bromine is denser than air (as the gas drifts over the road, forming a layer below air instead of rising), this is true – **bromine gas is denser than air.**

D is incorrect as the report suggests that bromine will dissolve in water to produce bromine water (as firemen were able to dissolve the gas), this is true – **bromine is soluble in water.** *Water induces a dipole in Br_2 enabling it to dissolve, however Br_2 dissolves more readily in organic solvents as it cannot form hydrogen bonds to water.*

3

The correct answer is **A** because:

- When concentrated sulfuric acid (H_2SO_4) reacts with a halogen it acts as both an **acid** and an **oxidising agent.**
  - Products formed due to H_2SO_4 acting as an acid: **hydrogen halide** and **sodium hydrogen sulfate.**
  - As H_2SO_4 acts as an oxidising agent the **halogen** is acting as a **reducing agent.**

- Only **bromine** and **iodine** are strong enough reducing agents to reduce the concentrated sulfuric acid (H_2SO_4).
  - For example, in potassium bromide; the bromide ions are oxidised to bromine.
  - The bromide ions reduce the sulfuric acid to sulfur dioxide gas.

- Therefore, a series of steps take place in the reaction of KBr and H_2SO_4:
  1. Sulfuric acid acts as an acid and donates a proton to the bromide ion:
     - $KBr + H_2SO_4 \rightarrow \mathbf{KHSO_4 + HBr}$
  2. The HBr gets oxidized and the sulfuric acid is reduced:
     - $2HBr + H_2SO_4 \rightarrow \mathbf{Br_2 + SO_2 + 2H_2O}$

- The complete list of products from both steps of this reaction are potassium hydrogensulfate ($KHSO_4$), hydrogen bromide (HBr), bromine (Br_2), sulfur dioxide (SO_2) and water (H_2O).

4

The correct answer is **C** because:

- The **thermal stability** of the hydrogen halides decreases as you go down the group. This is because the hydrogen halide bond is weaker.

- The bond gets weaker as you go down the group as the **halogen atoms are getting bigger** and the bond length is longer, and the bonding pair of electrons is getting further from the halogen nu*cl*eus.
- Therefore, hydrogen iodide will more readily decompose.

A is incorrect as high temperatures (450˚C) are needed to decompose ammonia.

B is incorrect as very high temperatures or electrolysis is required to split H_2O into its elements.

D is incorrect as hydrogen chloride will not decompose as readily as hydrogen iodide as the bond strength is larger.

5

The correct answer is **D** because:

- The reaction between chlorine and cold aqueous sodium hydroxide is:
  - $2NaOH + Cl_2 \rightarrow NaCl + NaClO + H_2O$
- The reaction between chlorine and hot aqueous sodium hydroxide is:
  - $6NaOH + 3Cl_2 \rightarrow 5NaCl + NaClO_3 + 3H_2O$
- So, both statements 1 and 2 are correct.
- A **disproportionation reaction** is a type of redox reaction. In this reaction, some of the atoms are reduced, and some are oxidised at the same in the same reaction.
- In cold alkali, chlorine is reduced to Na*Cl* and oxidised to Na*Cl*O.

5

The correct answer is **D** because:

- The reaction between chlorine and cold aqueous sodium hydroxide is:
  - $2NaOH + Cl_2 \rightarrow NaCl + NaClO + H_2O$
- The reaction between chlorine and hot aqueous sodium hydroxide is:
  - $6NaOH + 3Cl_2 \rightarrow 5NaCl + NaClO_3 + 3H_2O$
- So, both statements 1 and 2 are correct.
- A **disproportionation reaction** is a type of redox reaction. In this reaction, some of the atoms are reduced, and some are oxidised at the same in the same reaction.
- In cold alkali, chlorine is reduced to $NaCl$ and oxidised to $NaClO$.
- In hot alkali, chlorine is reduced to $NaCl$ and oxidised to $NaClO_3$.

6

The correct answer is **A** because:

- Chlorine is a diatomic molecule so each molecule of chlorine will have two atoms.
- Each of those atoms will have the same radius as the radius is determined by the number of electrons
- The molecular mass of the atom is determined by the number of neutrons in the nucleus.
- In the diatomic molecule, chlorine can have combinations of isotopes:
  - $^{35}Cl + {}^{35}Cl$, molecular mass = 70
  - $^{35}Cl + {}^{37}Cl$, molecular mass = 72
  - $^{37}Cl + {}^{37}Cl$, molecular mass = 74
  - As the molecular mass of this molecule is 72, it will have the $^{35}Cl + {}^{37}Cl$ atoms

B is incorrect as the isotopic mass is different between the two atoms.

C & **D** are incorrect as the nucleon number is the number of protons and neutrons. Since the number of neutrons is different in isotopes, the nucleon number will not be the same.

7

The correct answer is **B** because:

- In hot concentrated sulfuric acid, the astatide ion acts as a strong reducing agent.
- The sulfuric acid is reduced to hydrogen sulfide gas, and the astatide is oxidised to astatine.
- This is an example of a **redox** reaction:
  - $8NaAt\ (s) + 5H_2SO_4\ (l) \rightarrow 4Na_2SO_4\ (s) + 4At_2\ (s) + H_2\ (g) + 4H_2O\ (l)$
- As you go down the group the reducing power of the Group 17 atoms increases, or it is more easily oxidised.

A is incorrect as following the solubility trend that AgI is insoluble. AgAt will also be insoluble.

C is incorrect as a halogen atom can only displace a less reactive halide ion from its salt. Astatine is less reactive than chlorine so no reaction would take place.

D is incorrect as sulfur dioxide is also produced.

8

The correct answer is **A** because:

- There are only weak van der Waals' forces between the diatomic molecules, caused by instantaneous dipole-induced dipole forces.
- These forces increase as you go down Group 17 as the number of electrons in the molecules increase.
- The greater the number of electrons the greater the chance of instantaneous dipoles arising within molecules inducing dipoles in neighbouring molecules.
- The larger the molecules, the stronger the van der Waals' forces.
- Therefore, iodine has a stronger force than fluorine.

B is incorrect as the bond length increases in Group 17 as you go down the group.

C is incorrect as the bond energy decreases as you go down Group 17.

D is incorrect as the boiling point of Group 17 increases as you go down the group.

9

The correct answer is **B** because:

- The reaction produces sodium hypochlorite ($NaOCl$), this is used as a bleaching and disinfection agent and found in household bleach.
  - Statement 1 is correct.

- Silver nitrate will react with sodium chloride to produce a precipitate of silver chloride.
  - Statement 3 is correct.

Statement 2 is incorrect as the oxidation number of nitrogen in NCl_3 is +3; in N_2, it is 0. The change in oxidation number = 3 so nitrogen has gained 3 electrons, it has been reduced not oxidised.

10

The correct answer is **C** because:

- **Phosphorus trichloride**, PCl_3 is a simple covalent chloride, a liquid at room temperature.
  - It will not conduct electricity due to the lack of ions or mobile electrons.
  - It reacts violently with water to produce phosphorus acid H_3PO_3 and hydrogen chloride.